

Question 1

Suppose in a random experiment, there are two events A and B with the following known probabilities:

- $P(A) = 0.7$
- $P(B) = 0.6$
- $P(A \cap B) = 0.4$

Calculate the following probabilities:

1. $P(A \cup B)$: The probability that A or B occurs.
2. $P(\bar{A} \cap \bar{B})$: The probability that neither A nor B occurs.
3. $P(\bar{A} \cup \bar{B})$: The probability that A and B do not both occur.

Question 2 (Law of Total Probability)

A genome sequencing center uses three different high-throughput sequencers, which account for 40%, 35%, and 25% of all sequencing reads respectively.

The probabilities that a read contains a sequencing error on each platform are 0.015, 0.010, and 0.020, respectively.

If a sequencing read is chosen at random, what is the probability that it contains a sequencing error?

Question 3

A bag contains 1 red ball and 2 white balls. Each time a ball is drawn at random, its color is observed, then the ball is returned to the bag, and **3 additional balls of the same color as the drawn ball** are added to the bag. If we draw a ball from the bag three times in succession, find the probability that the first draw is a red ball, and the second and third draws are both white balls.

Question 4 (sampling with/without replacement).

In a controlled laboratory experiment investigating drug resistance, a researcher works with a finite culture volume containing a total of $N = 10$ bacterial colonies. Within this population, exactly $M = 6$ colonies exhibit resistance to a specific antibiotic. The researcher intends to randomly select a subsample of $n = 2$ colonies for phenotypic testing.

- (a) Assume the selection process involves sampling without replacement. Calculate the exact probability that the subsample contains exactly 1 resistant colony.
- (b) Suppose the researcher incorrectly treats the selection process as sampling with replacement, thereby assuming independence between selections. Calculate the resulting probability for observing exactly 1 resistant colony under this incorrect model.

Question 5 (Bayes' theorem).

OpenClaw is an AI agent deployed for real-time intrusion detection in a corporate network. It leverages a deep neural network to analyze network traffic and classify each packet as either "malicious" or "benign". Based on historical security logs, the probability that any given packet is malicious is 5%. The neural network has a true positive rate (sensitivity) of 98%—meaning it correctly identifies 98% of malicious packets—and a false positive rate of 3%—meaning it misclassifies 3% of benign packets as malicious. If OpenClaw processes a random packet and outputs a "malicious" classification, what is the probability that the packet is actually malicious?